



CSS Variables

https://www.w3schools.com/css/css3_variables.asp

Old way



```
body { background-color: #1e90ff; }
```

```
h2 { border-bottom: 2px solid #1e90ff; }
```

```
.container {  
  color: #1e90ff;  
  background-color: #ffffff;  
  padding: 15px;  
}
```

```
button {  
  background-color: #ffffff;  
  color: #1e90ff;  
  border: 1px solid #1e90ff;  
  padding: 5px;  
}
```

Old way



```
body { background-color: #1e90ff; } ←
```

```
h2 { border-bottom: 2px solid #1e90ff; } ←
```

```
.container {  
  color: #1e90ff; ←  
  background-color: #ffffff;  
  padding: 15px;  
}
```

```
button {  
  background-color: #ffffff;  
  color: #1e90ff; ←  
  border: 1px solid #1e90ff; ←  
  padding: 5px;  
}
```

Custom Properties



```
:root {  
  --blue: #1e90ff;  
  --white: #ffffff;  
}  
  
body { background-color: var(--blue); }  
  
h2 { border-bottom: 2px solid var(--blue); }  
  
.container {  
  color: var(--blue);  
  background-color: var(--white);  
  padding: 15px;  
}  
  
button {  
  background-color: var(--white);  
  color: var(--blue);  
  border: 1px solid var(--blue);  
  padding: 5px;  
}
```

Custom Properties



```
:root {  
  --blue: #1e90ff;  
  --white: #ffffff;  
}  
  
body { background-color: var(--blue); }  
  
h2 { border-bottom: 2px solid var(--blue); }  
  
.container {  
  color: var(--blue);  
  background-color: var(--white);  
  padding: 15px;  
}  
  
button {  
  background-color: var(--white);  
  color: var(--blue);  
  border: 1px solid var(--blue);  
  padding: 5px;  
}
```

Custom Properties



```
:root {  
  --blue: #1e90ff;  
  --white: #ffffff;  
}
```

```
body { background-color: var(--blue); }
```

```
h2 { border-bottom: 2px solid var(--blue); }
```

```
.container {  
  color: var(--blue);  
  background-color: var(--white);  
  padding: 15px;  
}
```

```
button {  
  background-color: var(--white);  
  color: var(--blue);  
  border: 1px solid var(--blue);  
  padding: 5px;  
}
```

var()



var(**<custom-property-name>** , **<declaration-value>?**)

```
:root {  
  --main-bg-color: pink;  
}  
  
body {  
  background-color: var(--main-bg-color);  
}
```

var()



var(**<custom-property-name>** , **<declaration-value>?**)

```
:root {  
  --main-bg-color: pink;  
}  
  
body {  
  background-color: var(--main-bg-color);  
}
```

```
/* Fallback */  
/* In the component's style: */  
.component .header {  
  color: var(--header-color, blue); /* header color */  
}  
  
.component .text {  
  color: var(--text-color, black);  
}
```



Global Variables

- Definite nell'elemento root

```
:root {  
  --blue: #6495ed;  
  --white: #faf0e6;  
}
```



Local Variables

```
.alert {  
  --alert-color: #ff6f69;  
}
```

La dichiaro in un selettore

```
.alert p {  
  color: var(--alert-color);  
  border: 1px solid var(--alert-color);  
}
```

La posso usare
nei descendant



Variabili e media query

- Posso riscrivere il valore di una variabile

```
:root {  
  --main-font-size: 16px;  
}  
  
media all and (max-width: 600px) {  
  :root {  
    --main-font-size: 12px;  
  }  
}
```



Grid Layout e Bootstrap



The New York Times

Monday, April 13, 2015 Today's Paper Video 81°F Dow -0.45%

World U.S. Politics New York Business Opinion Technology Science Health Sports Arts Style Food Home Travel Magazine Real Estate ALL

4 Ex-Employees of Blackwater Get Long Terms in Iraq Killings

By MATT APUZZO 59 minutes ago

One former Blackwater security contractor received a life sentence on Monday and three others received 30-year sentences for a shooting in Baghdad's Nisour Square in 2007, which left 17 dead.

Putin Lifts Ban on Sale of Air Defense Missiles to Iran

By NEIL MACFARQUHAR 36 minutes ago

President Vladimir V. Putin moved swiftly to take advantage of a possible thaw in relations due to a potential deal curbing Iran's nuclear ambitions.

Group Concludes U.S. Isn't Following Its Own Drone Rules

By SCOTT SHANE 4:03 PM ET

A study of drone strikes in Yemen concludes that the Obama administration has not followed its own rules to avoid civilian casualties.

Immelt Is Putting His Own Stamp on Jack Welch's G.E.

ILLUSTRATION BY KYLE CLARK



Senator Marco Rubio of Florida will make a formal announcement Monday evening in Miami. Joe Buglewicz for The New York Times

Rubio Announces Presidential Bid

By ASHLEY PARKER

Senator Marco Rubio of Florida told his top donors Monday that he is running for president in 2016, becoming the third Republican to officially enter the contest.

MORE ON MARCO RUBIO

- The Upshot: Political Skill May Not Be Enough for Rubio
- Path to the Presidency | On the Issues | Things You May Not Know

MORE ON ELECTION 2016

- From the Magazine: It's Big Announcement Season

Volunteer Deputy Charged With Manslaughter in Tulsa

By RICHARD PÉREZ-PEÑA 31 minutes ago

Robert Charles Bates, 73, a reserve deputy with the Tulsa County Sheriff's Department in Oklahoma, fatally shot a suspect.

GÜNTER GRASS | 1927-2015

Writer Pried Open Germany's Past but Hid His Own

By STEPHEN KINZER

Mr. Grass was called Germany's moral conscience by many, but he stunned Europe when he revealed in 2006 that he had been a member of the Waffen-SS during World War II.

173 Comments

- ArtsBeat: Looking Back at Günter Grass's Life and Work



The Opinion Pages

New York's Leaky Public Pension Funds

By THE EDITORIAL BOARD

The city's comptroller needs to come up with better ways to protect taxpayer funds.

- Editorial: 152 Innocents, Marked for Death
- Blow: Walter Scott Is Not on Trial
- Krugman: It Takes a Party
- Room for Debate: Bringing Up Baby on the iPhone?
- Opinion on Facebook »

The Power of Light

He lowered the magnifying glass over the ant, intensifying the assault, cornering his prey.



Philosophy Returns to the Real World

I ran out of interest in my own consciousness, but there's no reason ever to run out of interest in the world.

Watching

33m The F.B.I. is investigating accusations that San Francisco County deputies forced inmates to **engage in gladiator-style fights** for gambling and entertainment purposes.
The Los Angeles Times »

40m **SpaceX** @SpaceX Follow
Launch scrubbed today for weather, next opportunity tomorrow 4/14 at 4:10pm EDT.
SpaceX, on the delay of its third attempt to land a used rocket on a barge in the ocean, via Twitter »

1h A white reserve sheriff's deputy who fatally shot a black man during an arrest was **charged with manslaughter** in Oklahoma. The officer said he had meant to use his Taser.
Tulsa World »

1h Two former **egg industry executives**, a father and son, were sentenced to three months in jail after a nationwide salmonella outbreak in 2010 caused by contaminated eggs.
The Associated Press »

1h New measurements by NASA's Curiosity rover have led scientists to believe they have found **the first evidence of liquid water on Mars**.



SECTIONS

SEARCH

U.S.INTERNATIONAL中文

The New York Times

Monday, April 13, 2015Today's PaperVideo81°FMoscow -43.158%

WorldU.S. Politics New York Business Opinion Technology Science Health Sports Arts Style Food Home Travel Magazine Real Estate ALL

4 Ex-Employees of Blackwater Get Long Terms in Iraq Killings

By MATT APUZZO 59 minutes ago

One former Blackwater security contractor received a life sentence on Monday and three others received 30-year sentences for a shooting in Baghdad's Nisour Square in 2007, which left 17 dead.

Putin Lifts Ban on Sale of Air Defense Missiles to Iran

By NEIL MACFARQUHAR 35 minutes ago

President Vladimir V. Putin moved swiftly to take advantage of a possible thaw in relations due to a potential deal curbing Iran's nuclear ambitions.

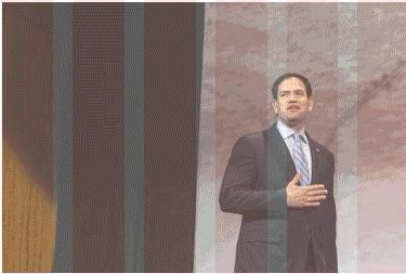
Group Concludes U.S. Isn't Following Its Own Drone Rules

By SCOTT SHANE 4:03 PM ET

A study of drone strikes in Yemen concludes that the Obama administration has not followed its own rules to avoid civilian casualties.

Immelt Is Putting His Own Stamp on Jack Welch's G.E.

By SCOTT SHANE 4:03 PM ET



Senator Marco Rubio of Florida will make a formal announcement Monday evening in Miami. Joe Bugiewicz for The New York Times

Rubio Announces Presidential Bid

By ASHLEY PARKER

Senator Marco Rubio of Florida told his top donors Monday that he is running for president in 2016, becoming the third Republican to officially enter the contest.

MORE ON MARCO RUBIO

- The Upshot: Political Skill May Not Be Enough for Rubio
- Path to the Presidency | On the Issues | Things You May Not Know

MORE ON ELECTION 2016

- From the Magazine: It's Big Announcement Season

Volunteer Deputy Charged With Manslaughter in Tulsa

By RICHARD PÉREZ-PÉÑA 35 minutes ago

Robert Charles Bates, 73, a reserve deputy with the Tulsa County Sheriff's Department in Oklahoma, fatally shot a suspect.

GÜNTER GRASS | 1927-2015

Writer Pried Open Germany's Past but Hid His Own

By STEPHEN KINZER

Mr. Grass was called Germany's moral conscience by many, but he stunned Europe when he revealed in 2006 that he had been a member of the Waffen-SS during World War II.

173 Comments

- ArtsBeat: Looking Back at Günter Grass's Life and Work

New York's Leaky Public Pension Funds

By THE EDITORIAL BOARD

The city's comptroller needs to come up with better ways to protect taxpayer funds.

- Editorial: 152 Innocents, Marked for Death
- Blow: Walter Scott Is Not on Trial
- Krugman: It Takes a Party
- Room for Debate: Bringing Up Baby on the iPhone?
- Opinion on Facebook »

The Opinion Pages

MENAGERIE | PAMELA GWYN KRIPKE

The Power of Light

He lowered the magnifying glass over the ant, intensifying the assault, cornering his prey.

THE STONE | CRISPIN SARTWELL

Philosophy Returns to the Real World

I ran out of interest in my own consciousness, but there's no reason ever to run out of interest in the world.

Watching

32m The F.B.I. is investigating accusations that San Francisco County deputies forced inmates to **engage in gladiator-style fights** for gambling and entertainment purposes. [The Los Angeles Times »](#)

40m  **SpaceX** @SpaceX [Follow](#)

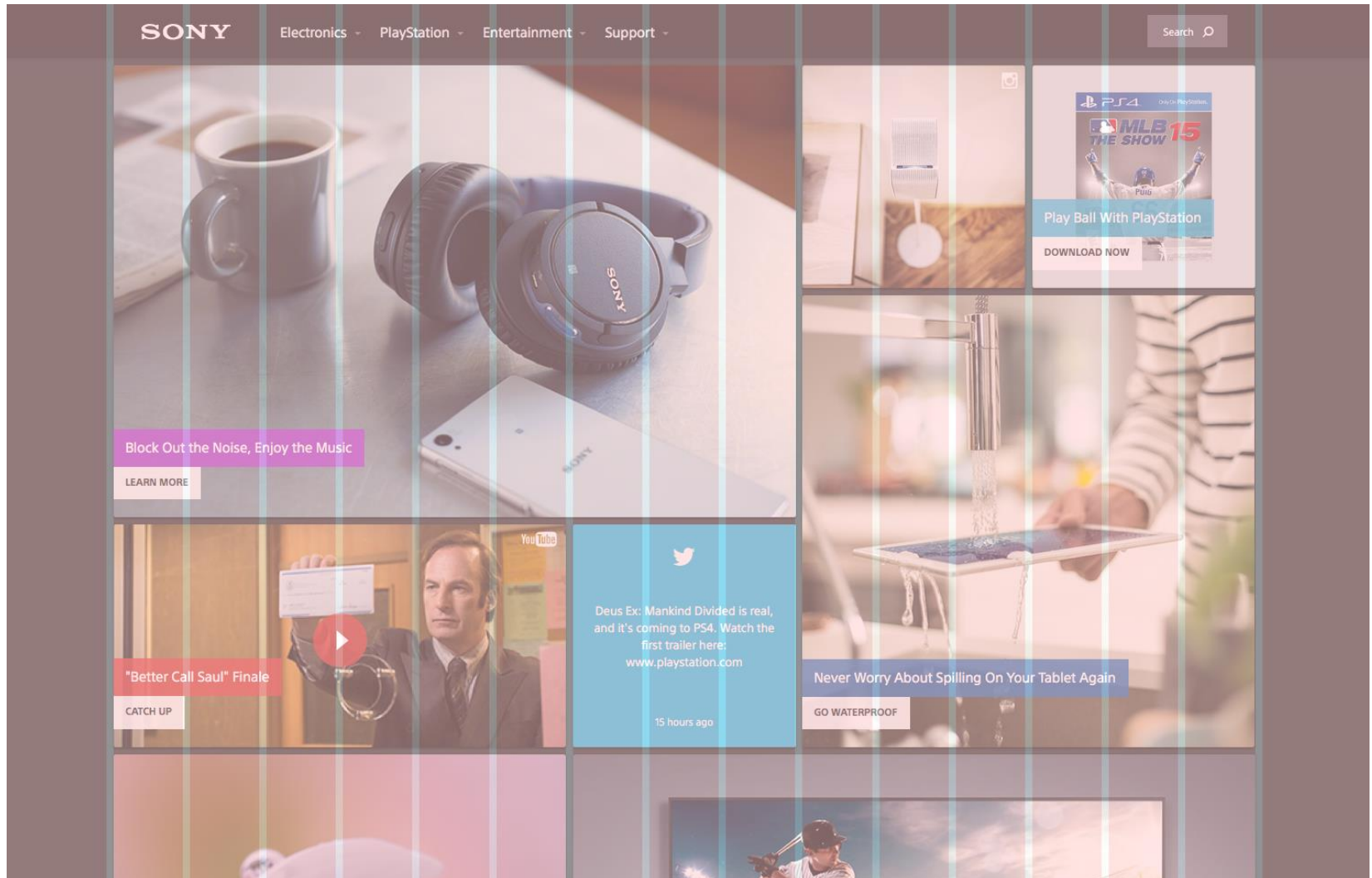
Launch scrubbed today for weather, next opportunity tomorrow 4/14 at 4:10pm EDT.

SpaceX, on the delay of its third attempt to land a used rocket on a barge in the ocean, via Twitter »

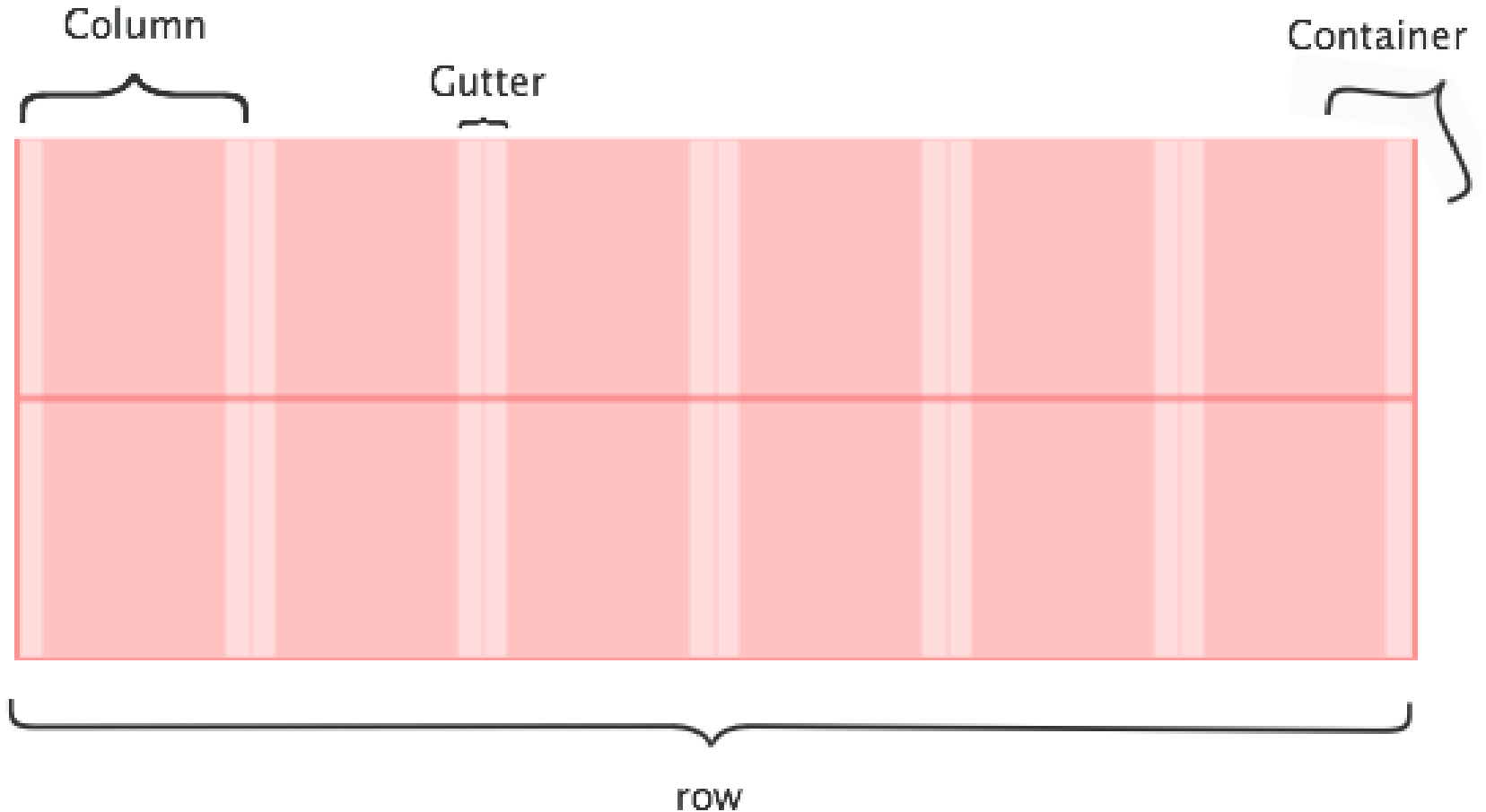
1h A white reserve sheriff's deputy who fatally shot a black man during an arrest was **charged with manslaughter** in Oklahoma. The officer said he had meant to use his Taser. [Tulsa World »](#)

1h Two former **egg industry executives**, a father and son, were sentenced to three months in jail after a nationwide salmonella outbreak in 2010 caused by contaminated eggs. [The Associated Press »](#)

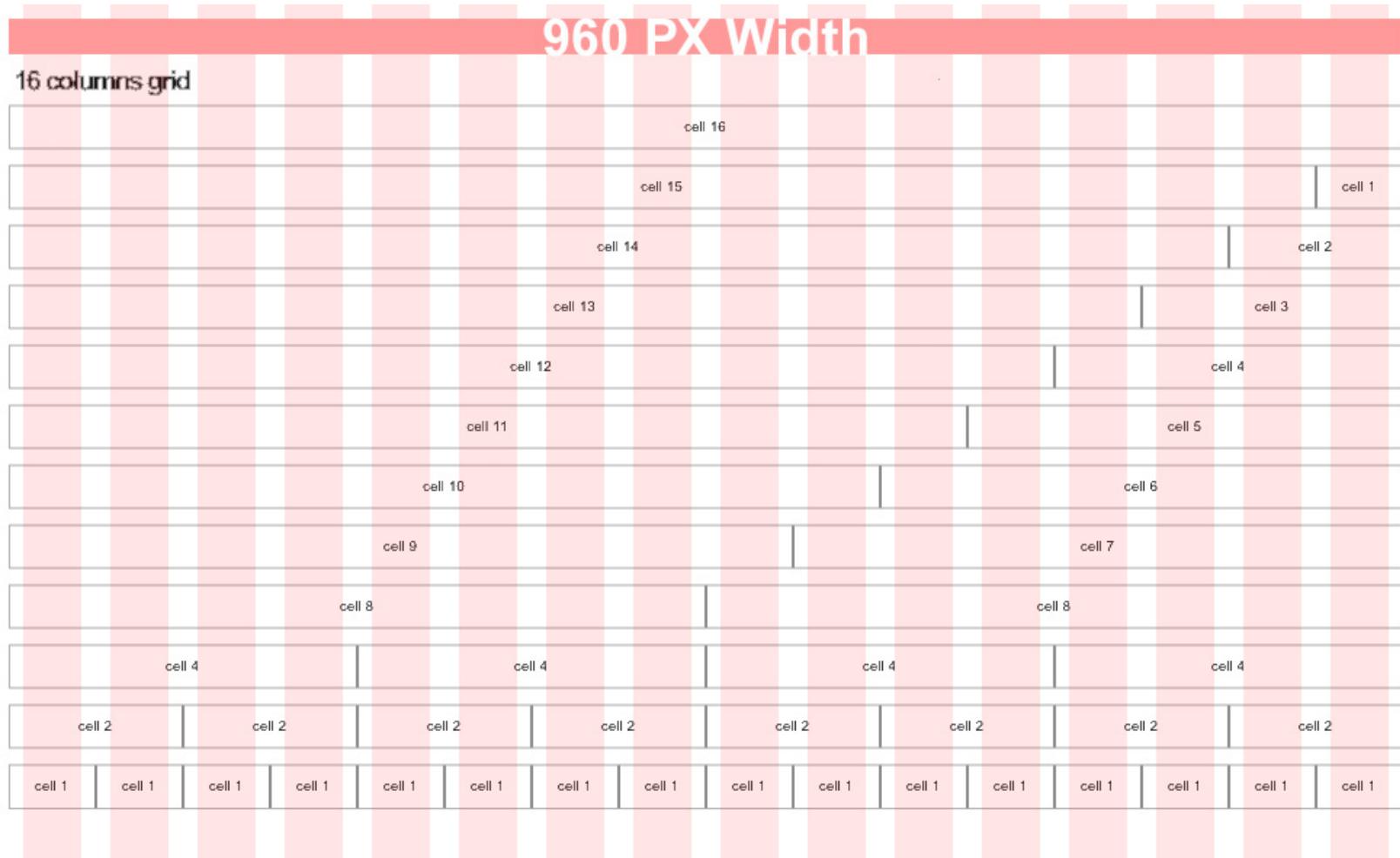
1h New measurements by NASA's Curiosity rover have led scientists to believe they have found **the first evidence of liquid water on Mars**. [The New York Times »](#)



Struttura



Grid system





12 Column Grid

940											
60	860										
140		780									
220			700								
300				620							
380					540						
460						460					
220			220			60	380				
60	380				220			220			

60

140

220

300

380

460

220

60

60

220

220



Framework Responsive

- Bootstrap

span 1	span 1	span 1	span 1	span 1	span 1	span 1	span 1	span 1	span 1	span 1	span 1
span 4				span 4				span 4			
span 4				span 8							
span 6						span 6					
span 12											

- W3.CSS

1/2		1/2		1/3		1/3		1/3		1/3		2/3																					
1/4	1/4			1/4	1/4			1/4			1/2			1/4	1/4		2/3			1/3													
3/4					1/4		1/4		1/4		1/2			1/4		1/4		1/2		1/4													
50px	rest											1/4		rest											100px		rest					45px	



Grid system

CSS grid



- **Grid** is a two-dimensional system,
 - handle columns and rows,
 - **flexbox** a one-dimensional system

Two dimensions



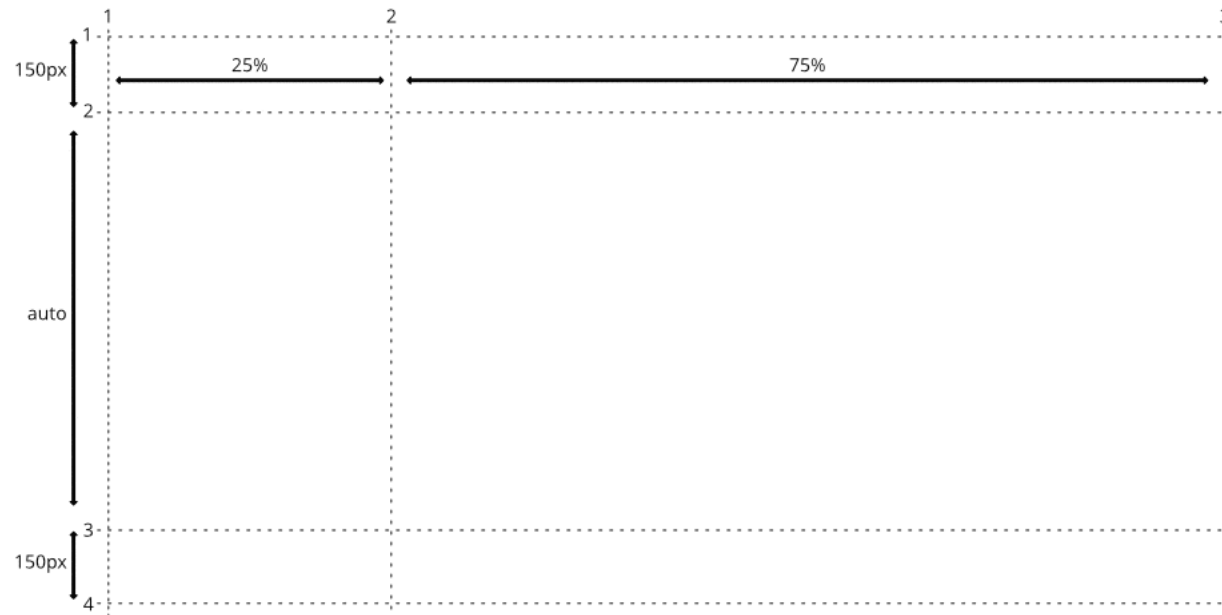
One dimension



[can i use grid?](https://caniuse.com/)
<https://caniuse.com/>

grid lines



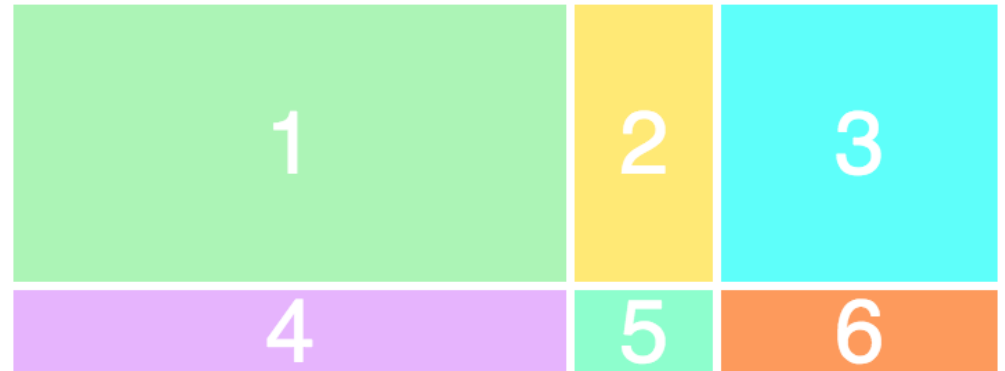


grid template



```
.wrapper {  
  display: grid;  
  grid-template-columns: 200px 50px 100px;  
  grid-template-rows: 100px 30px;  
}
```

```
<div class="wrapper">  
  <div>1</div>  
  <div>2</div>  
  <div>3</div>  
  <div>4</div>  
  <div>5</div>  
  <div>6</div>  
</div>
```



<https://www.freecodecamp.org/news/learn-css-grid-in-5-minutes-f582e87b1228/>



Esempi di grid template

grid-template-rows: 320px auto 320px;

grid-template-rows: 250px 800px 400px;

grid-template-rows: repeat(3, 275px);

grid-template-rows: 1fr 2fr 1fr;

grid-template-columns: 25% 75%;

grid-template-columns: 100px 100px 100px 100px;

grid-template-columns: repeat(4, 100px);

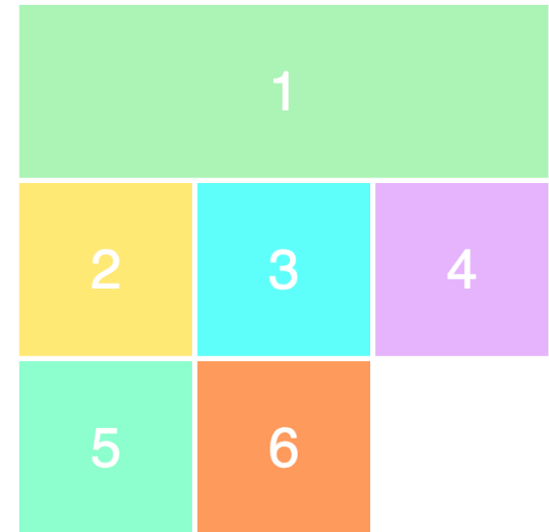
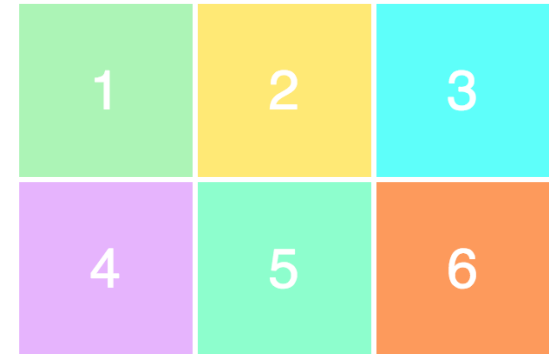
grid-template-columns: 200px auto minmax(80px, auto)

Posizionamento item



```
.wrapper {  
  display: grid;  
  grid-template-columns: 100px 100px 100px;  
  grid-template-rows: 100px 100px 100px;  
}
```

```
.item1 {  
  grid-column-start: 1;  
  grid-column-end: 4;  
}
```

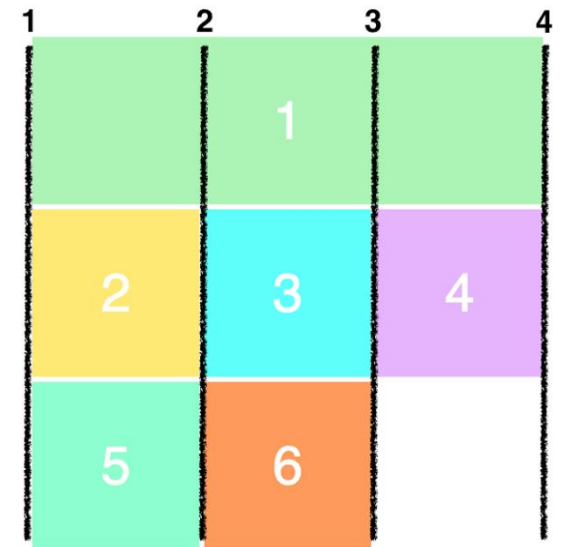
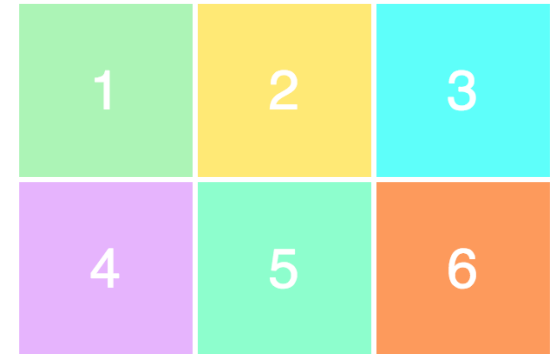


Posizionamento item



```
.wrapper {  
  display: grid;  
  grid-template-columns: 100px 100px 100px;  
  grid-template-rows: 100px 100px 100px;  
}
```

```
.item1 {  
  grid-column-start: 1;  
  grid-column-end: 4;  
}
```

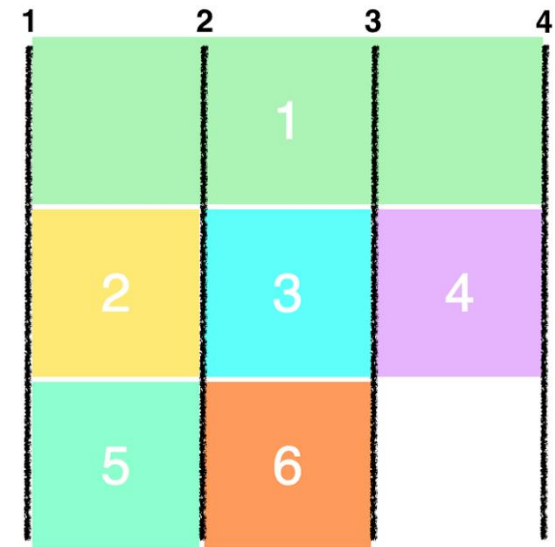
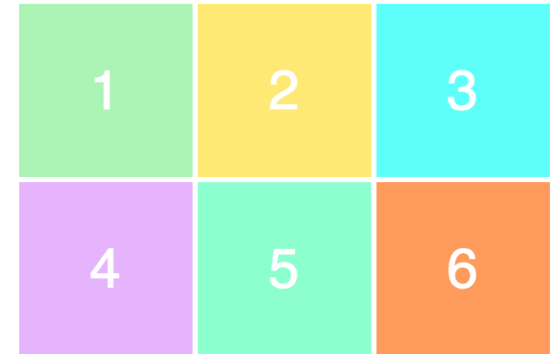


Posizionamento item



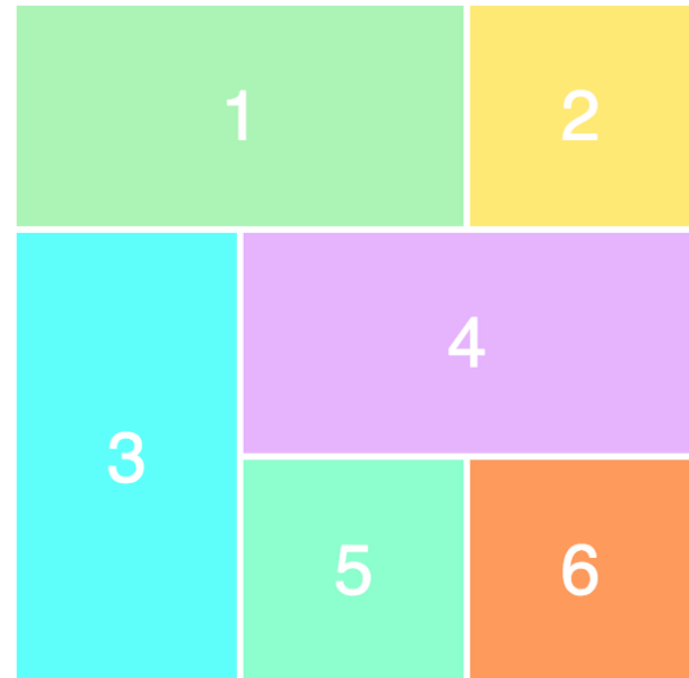
```
.wrapper {  
  display: grid;  
  grid-template-columns: 100px 100px 100px;  
  grid-template-rows: 100px 100px 100px;  
}
```

```
.item1 {  
  grid-column: 1/4;  
}
```





```
.item1 {  
    grid-column-start: 1;  
    grid-column-end: 3;  
}  
  
.item3 {  
    grid-row-start: 2;  
    grid-row-end: 4;  
}  
  
.item4 {  
    grid-column-start: 2;  
    grid-column-end: 4;  
}
```



Page Layout



```
<div class="wrapper">
  <header>HEADER</header>
  <nav>MENU</nav>
  <main>CONTENT</main>
  <footer>FOOTER</footer>
</div>
```

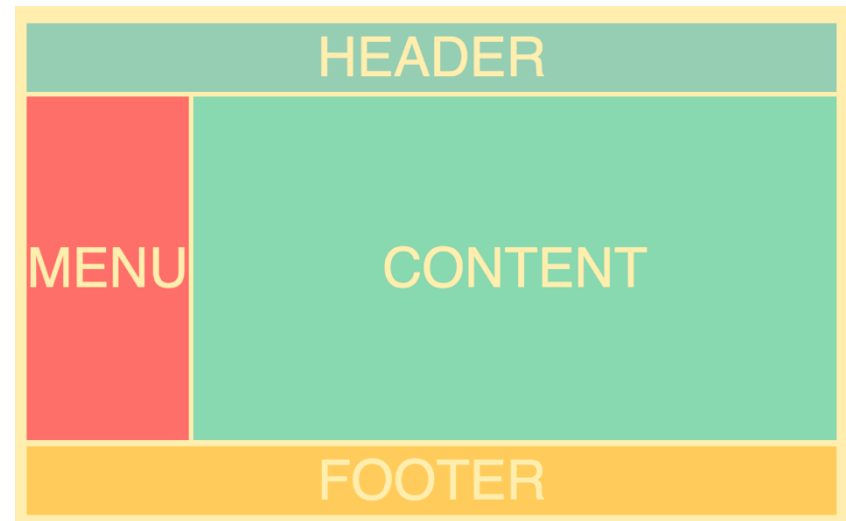


```
.wrapper{
  display: grid;
  grid-gap: 3px;
  grid-template-columns: repeat(12, 1fr);
  grid-template-rows: 40px 200px 40px;
}
```



```
.wrapper{  
  display: grid;  
  grid-gap: 3px;  
  grid-template-columns: repeat(12, 1fr);  
  grid-template-rows: 40px 200px 40px;  
}
```

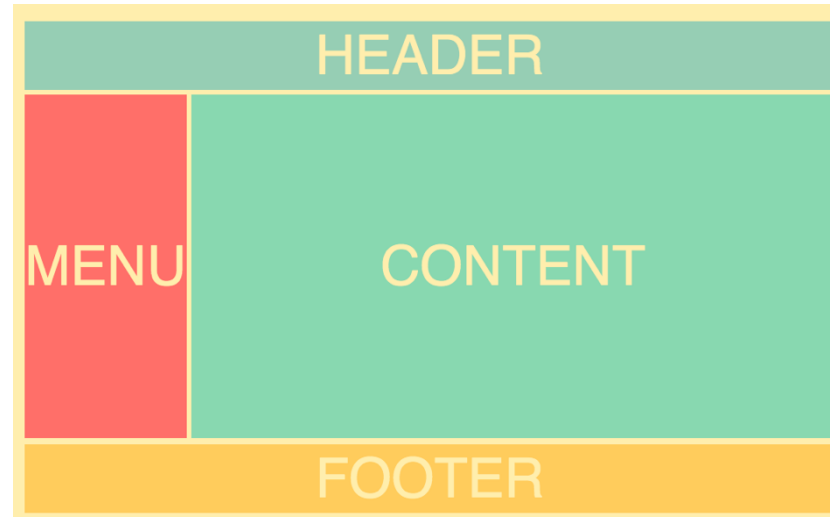
```
header{ grid-column: 1/ -1;}  
nav{ }  
main{   grid-column: 2/ -1;}  
footer{ grid-column: 1 / -1;}
```





```
.wrapper{  
  display: grid;  
  grid-gap: 3px;  
  grid-template-columns: repeat(12, 1fr);  
  grid-template-rows: 40px 200px 40px;  
  grid-template-areas: "h h h h h h h h h h h h"  
                      "m c c c c c c c c c c c"  
                      "f f f f f f f f f f f f";  
}
```

```
header{ grid-area: h;}  
nav{    grid-area: m;}  
main{   grid-area: c;}  
footer{ grid-area: f;}
```





GRID FRAMEWORKS

Bootstrap



- **Easy to use**
 - Anybody with just basic knowledge of HTML and CSS can start using Bootstrap
- **Responsive features**
 - Bootstrap's responsive CSS adjusts to phones, tablets, and desktops
- **Mobile-first approach**
 - In Bootstrap 3, mobile-first styles are part of the core framework
- **Browser compatibility**
 - Bootstrap is compatible with all modern browsers (Chrome, Firefox, Internet Explorer, Safari, and Opera)

Bootstrap grid



	xs <576px	sm ≥576px	md ≥768px	lg ≥992px	xl ≥1200px	xxl ≥1400px
Container <i>max-width</i>	None (auto)	540px	720px	960px	1140px	1320px
Class prefix	<i>.col-</i>	<i>.col-sm-</i>	<i>.col-md-</i>	<i>.col-lg-</i>	<i>.col-xl-</i>	<i>.col-xxl-</i>
# of columns	12					
Gutter width	1.5rem (.75rem on left and right)					
Custom gutters	Yes					
Nestable	Yes					
Column ordering	Yes					

<https://getbootstrap.com/>



Esempio grid 1

```
<div class="container">
  <h1>Hello World!</h1>
  <div class="row">
    <div class="col-sm-6" style="background-color:yellow;">
      <p>Lorem ipsum...</p>
    </div>
    <div class="col-sm-6" style="background-color:pink;">
      <p>Sed ut perspiciatis...</p>
    </div>
  </div>
</div>
```

span 6

span 6

span 6

span 6



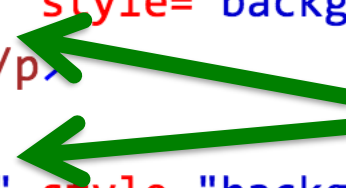
Esempio grid 2

```
<div class="container-fluid"> ←  
  <h1>Hello World!</h1>  
  <div class="row">  
    <div class="col-sm-6" style="background-color:yellow;">  
      <p>Lorem ipsum...</p>  
    </div>  
    <div class="col-sm-6" style="background-color:pink;">  
      <p>Sed ut perspiciatis...</p>  
    </div>  
  </div>  
</div>
```



Esempio grid 3

```
<div class="container-fluid">
  <h1>Hello World!</h1>
  <div class="row">
    <div class="col-sm-3" style="background-color:yellow;">
      <p>Lorem ipsum...</p>
    </div>
    <div class="col-sm-9" style="background-color:pink;">
      <p>Sed ut perspiciatis...</p>
    </div>
  </div>
</div>
```



http://www.w3schools.com/bootstrap/tryit.asp?filename=trybs_grid_small&stacked=h

Esempio grid 4



```
<div class="container-fluid">
  <h1>Hello World!</h1>
  <div class="row">
    <div class="col-sm-3 col-md-6" style="background-color:yellow;">
      <p>Lorem ipsum...</p>
    </div>
    <div class="col-sm-9 col-md-6" style="background-color:pink;">
      <p>Sed ut perspiciatis...</p>
    </div>
  </div>
</div>
```



- Esempi Bootstrap
- <http://startbootstrap.com>